BASIC FLIGHT CONTROLLER – GYROSCOPE, ACCELEROMETER, DISTANCE SENSOR

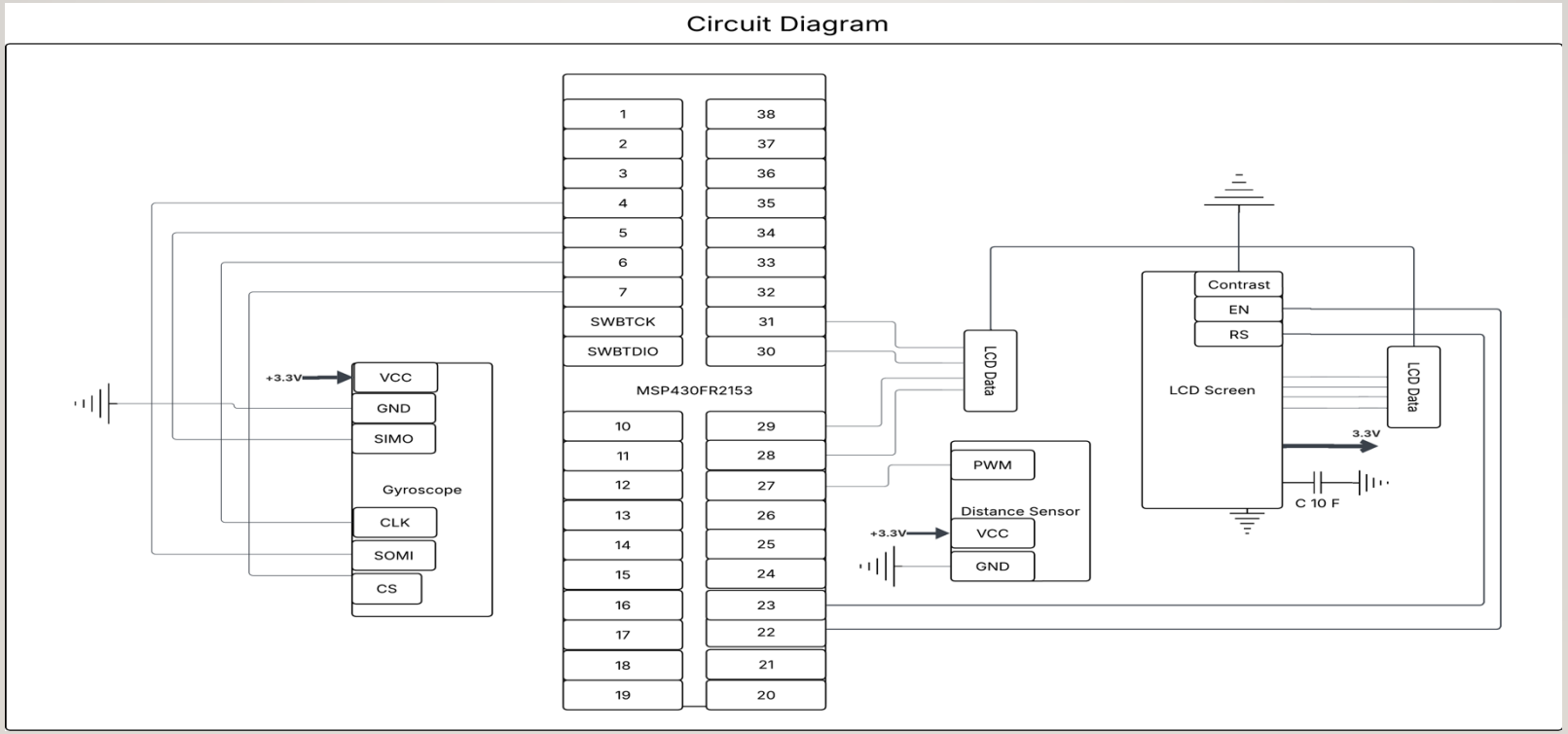
EELE 465 MICROCONTROLLER APPLICATIONS FINAL PROJECT

RYAN ADOLPH

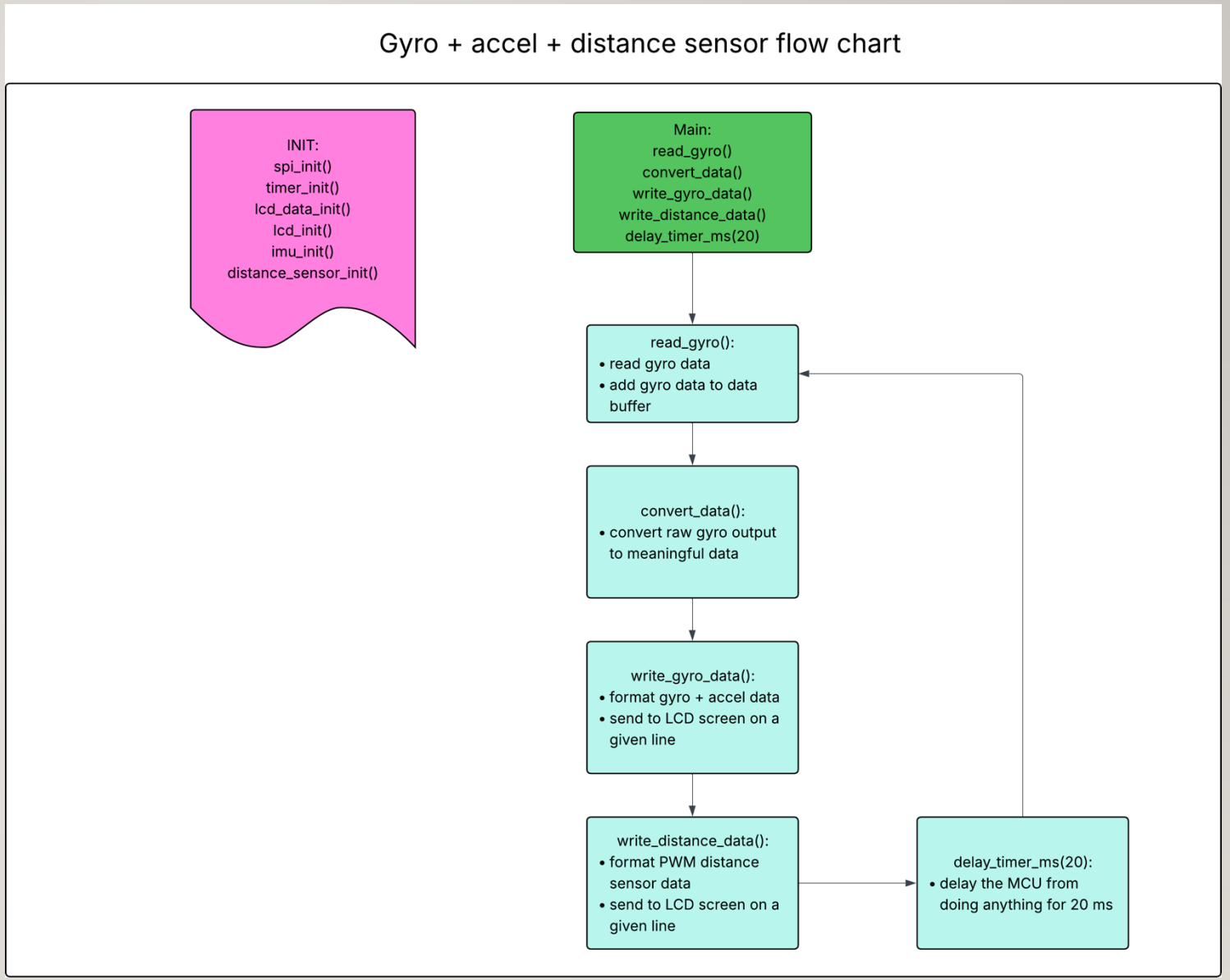
GOAL

- The goal of this project was the be the basis for a future drone flight controller
- I wanted to be able to use a gyroscope to detect my boards 3-D acceleration and change in position so my MCU will have some information of where it is in space
- I also wanted to implement a distance sensor as the start of a self-landing + takeoff vehicle that is aware of how close it is to the ground or other objects
- Output these values onto an LCD display so that the user is able to read the data in real time

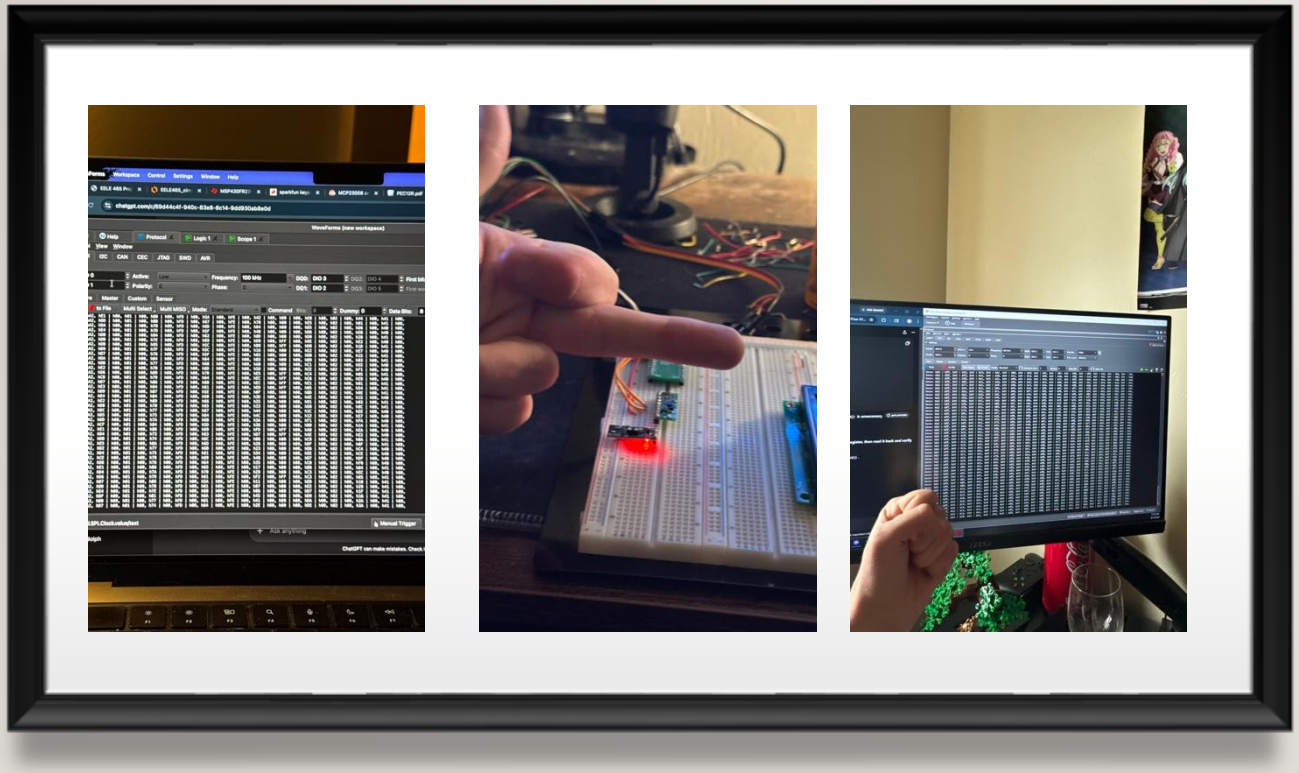
CIRCUIT DIAGRAM



MSP430FR2153 MAIN FLOWCHART



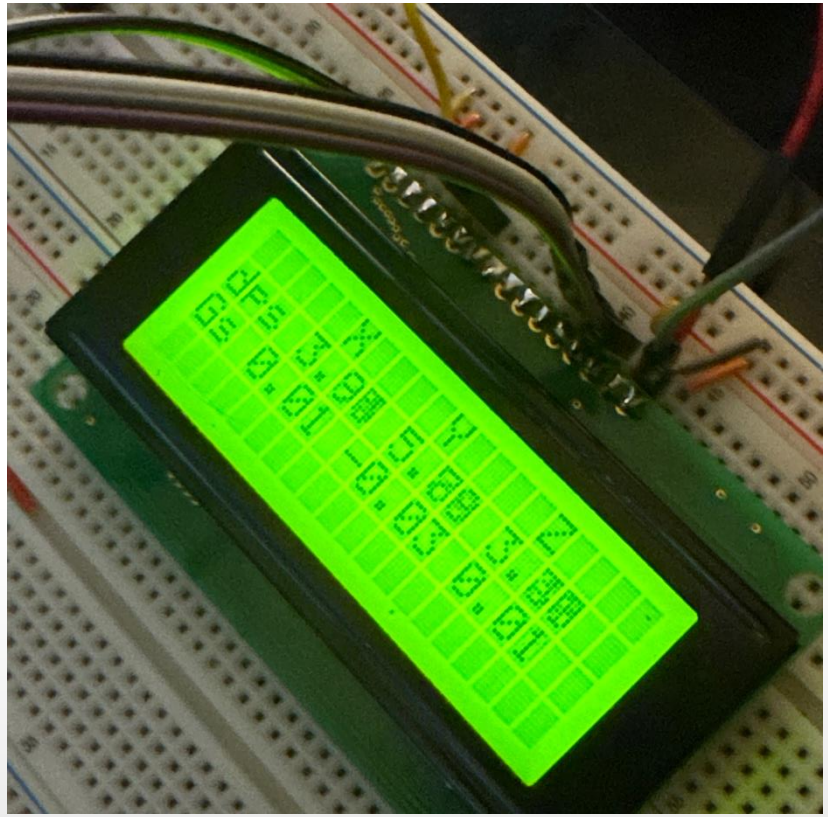
DEVELOPMENT
APRIL 6TH – APRIL 16TH

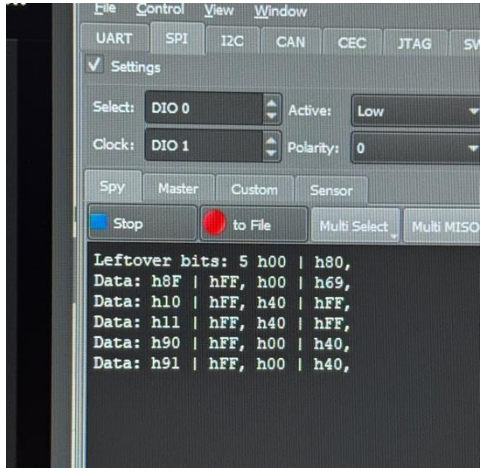


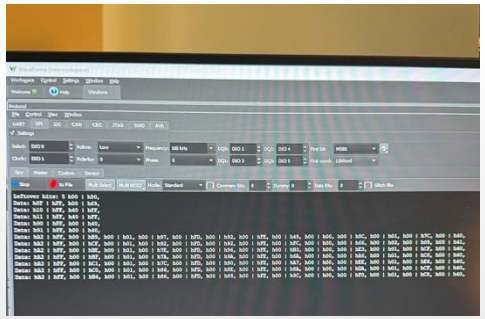
- Left: April 6th – got the gyroscope initially programmed and outputting something
- Right: April 10th – gyro is not outputting the correct data
- Middle: April 10th – I gave up on getting it working for this week

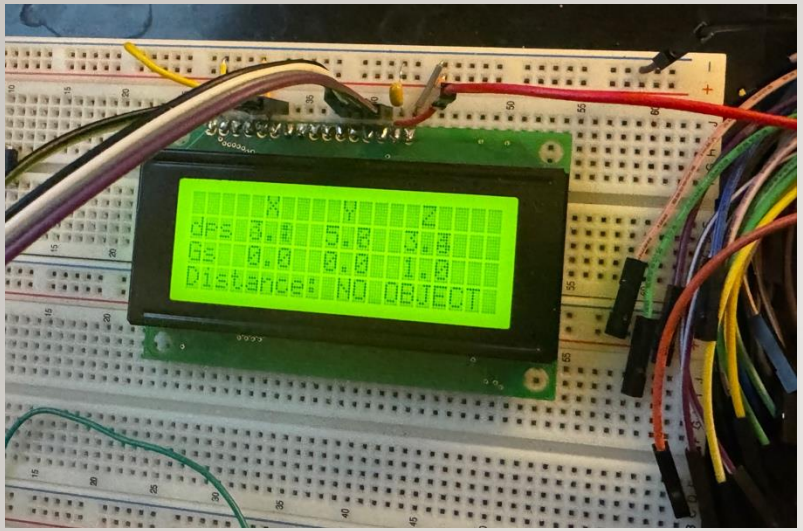
DEVELOPMENT CONT.

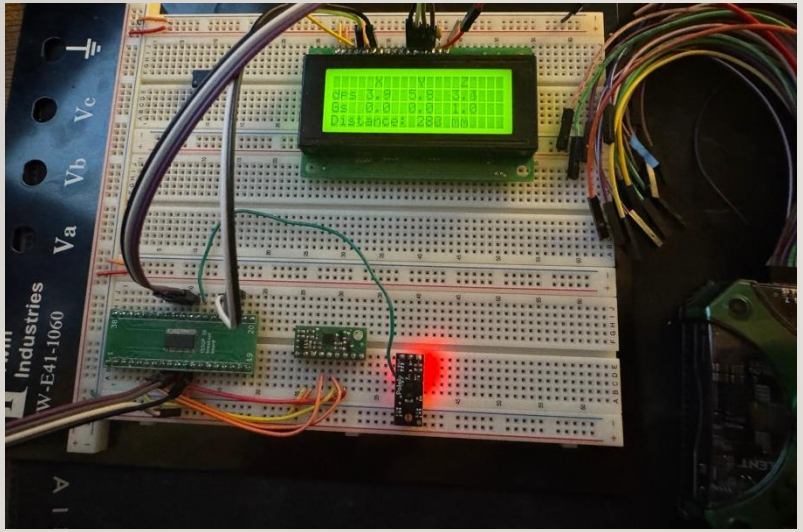
- Left: April 14th – got the gyro working!! Also learned I soldered it upside-down
- Right: April 14th – finally have the gyroscope outputting correct values from debugging + reading its data











DEVELOPMENT END

- Bottom: April 16th – Got the PWM distance sensor implemented
- Top: April 16th – Implemented Too-Close and Too-Far / No Object detected printing for distance sensor. Need to test its accuracy and adjust distance calculations